Task Scheduling Project Comparison of the results:

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Calculation of the Throughputs:

Throughput = Number of Cloudlets / Total Finish Time

- For FCFS: Throughput = 10/ 366.76 = 0.0272
- For Shortest Remaining Time: Throughput = 10 / 260.10 = 0.0384
- For Min-Min: Throughput = 10 / 237.60 = 0.0421
- For Max-Min: Throughput = 10 / 218.43 = 0.0458
- For Sufferage: Throughput = 10 / 230.10 = 0.0435

Calculation of the Turnaround Time:

Average Turnaround = Sum of Finish Times / Number of Cloudlets

- For FCFS: Turnaround Time = (100.1 + 112.6 + 133.43 + 133.54 + 142.6 + 153.48 + 190.1 + 213.43 + 306.76 + 366.76) / 10 = 185.28

- For Shortest Remaining Time: Turnaround Time = (66.77 + 91.76 + 141.76 + 157.6 + 161.76 + 180.09 + 193.43 + 212.59 + 257.6 + 260.1 / 10 = 172.35

- For Min-Min: Turnaround Time = (83.43 + 112.59 + 133.43 + 136.77 + 151.76 + 187.59 + 188.43 + 200.1 +205.1 + 237.6) / 10 = 163.68

- For Max-Min: Turnaround Time = (83.43 + 112.59 + 133.43 + 136.77 + 151.77 + 187.59 + 188.43 + 200.1 + 217.6 + 218.43) / 10 = 163.01

- For Sufferage: Turnaround Time = (133.43 + 150.1 + 160.1 + 183.43 + 186.76 + 200.1 + 203.43 + 203.54 + 213.43 + 230.1) / 10 = 186.44

Algorithm	Throughput	Average Turnaround Time
FCFS	0.0272	185.28
Shortest Remaining Time	0.0384	172.35
Min-Min	0.0421	163.68
Max-Min	0.0458	163.01
Sufferage	0.0435	186.44

From all the results, FCFS performed the worst overall, with the lowest throughput of 0.0272 and a high turnaround time. This is because FCFS just runs tasks in order and does not consider any task sizes or VM speed, which makes tasks wait a long time to process. On the other hand, Max-Min is the best algorithm in this experiment. Achieving a throughput of 0.0458, an average Turnaround Time of 163.01. Because it schedules bigger tasks first in a smart way, it balances the load between VMs and reduces total execution time.